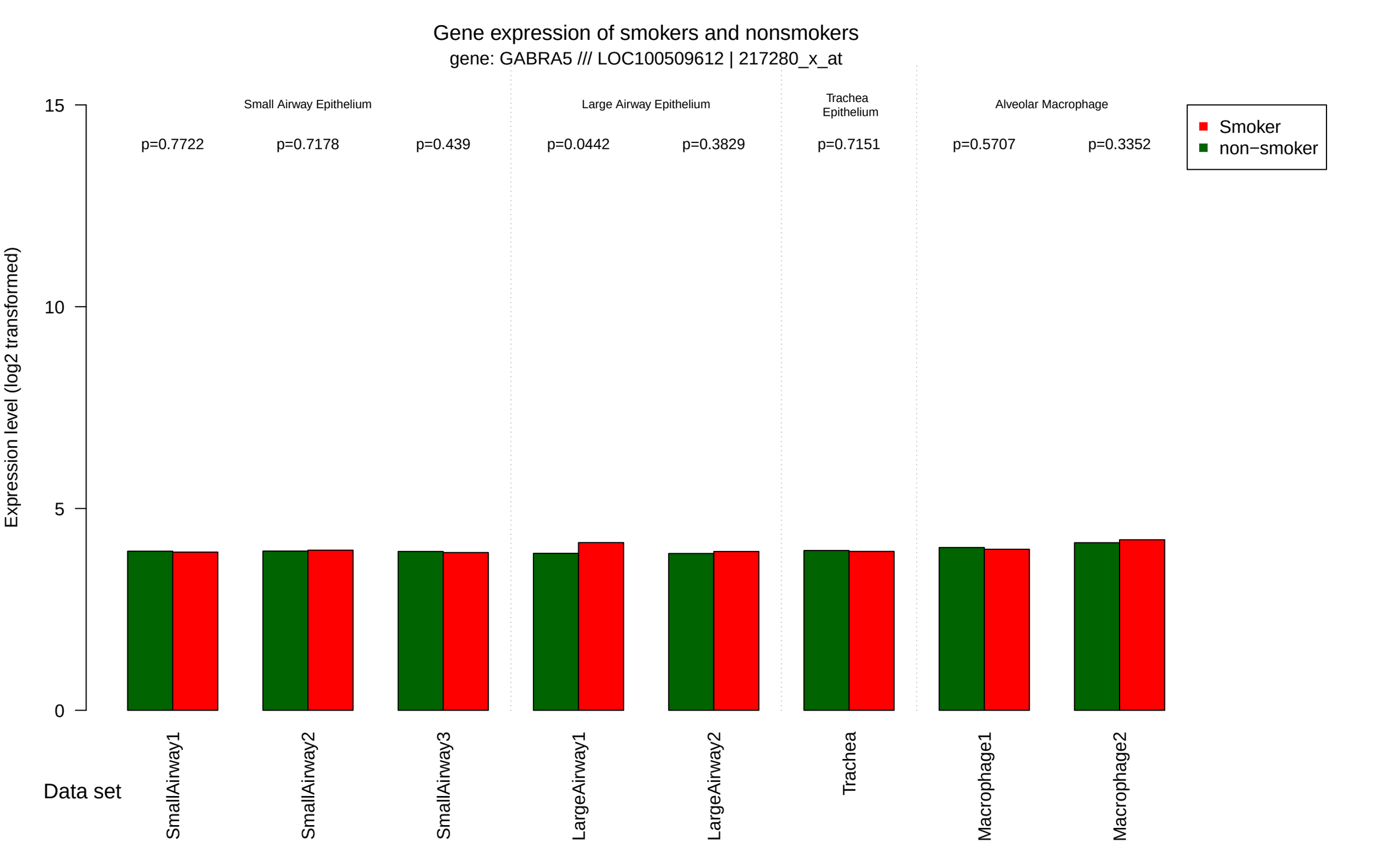
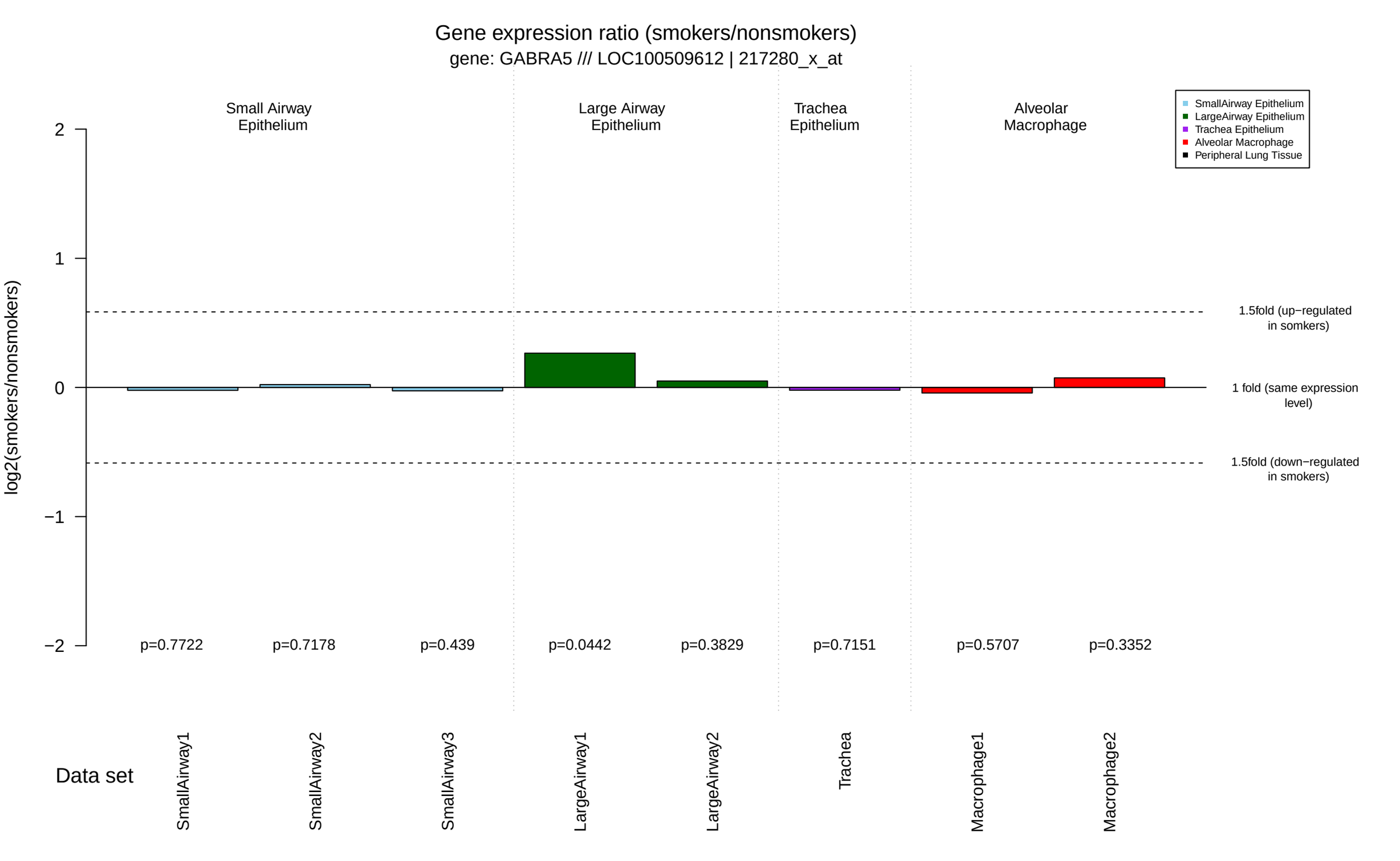
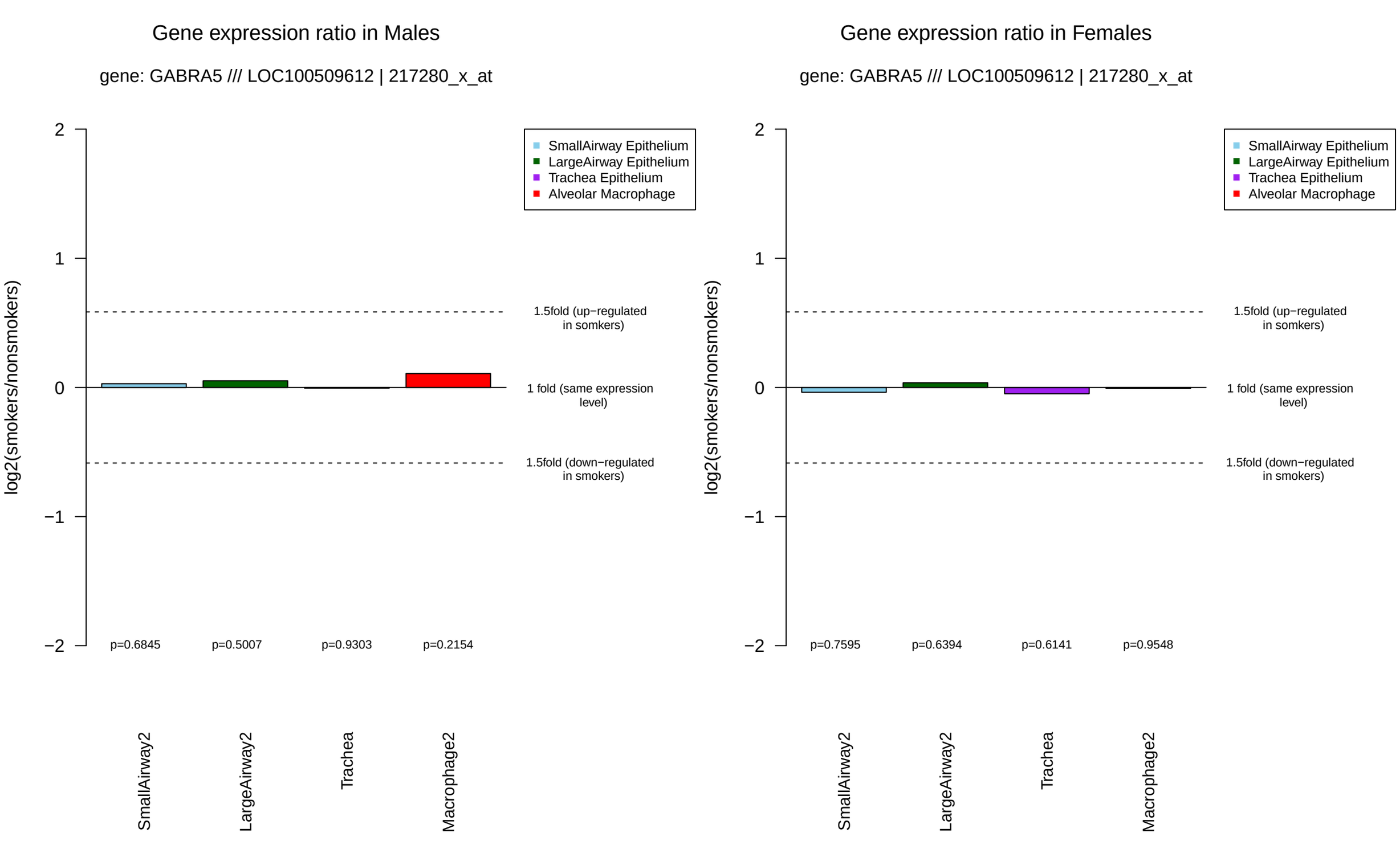
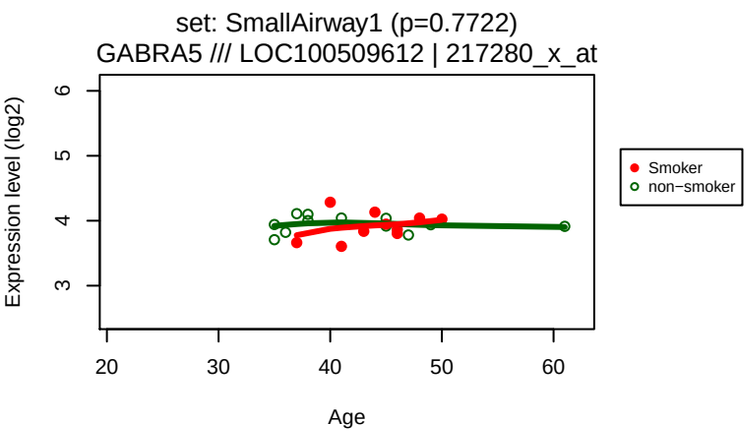
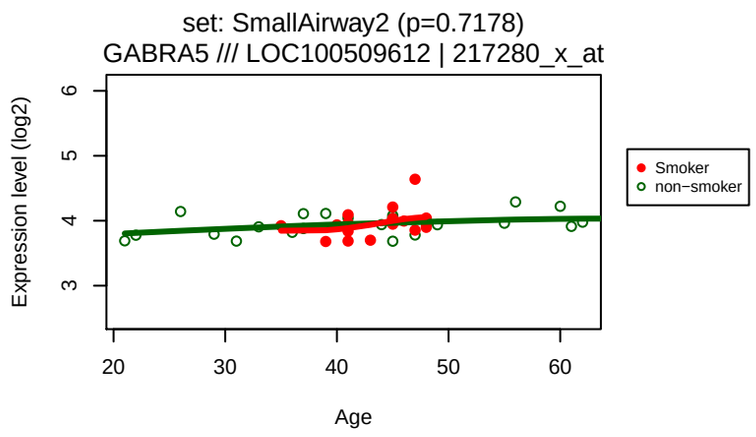
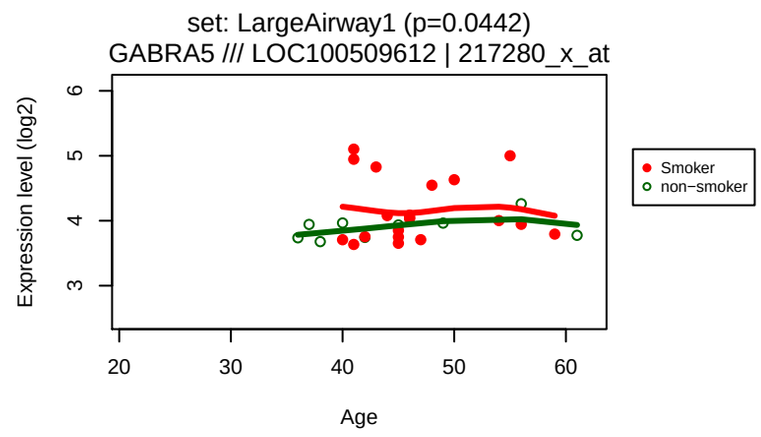

Gene expression of smokers and nonsmokers
gene: GABRA5 /// LOC100509612 | 217280_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7722
p=0.7178
p=0.439
p=0.0442
p=0.3829
p=0.7151
p=0.5707
p=0.3352
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

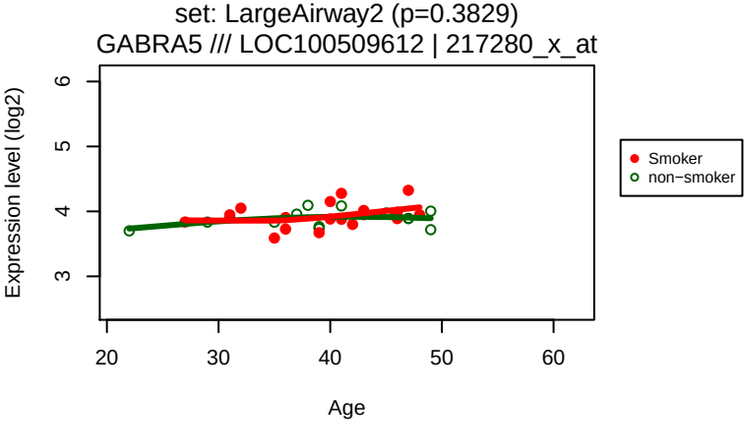

Gene expression ratio (smokers/nonsmokers)
gene: GABRA5 /// LOC100509612 | 217280_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7722
p=0.7178
p=0.439
p=0.0442
p=0.3829
p=0.7151
p=0.5707
p=0.3352
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

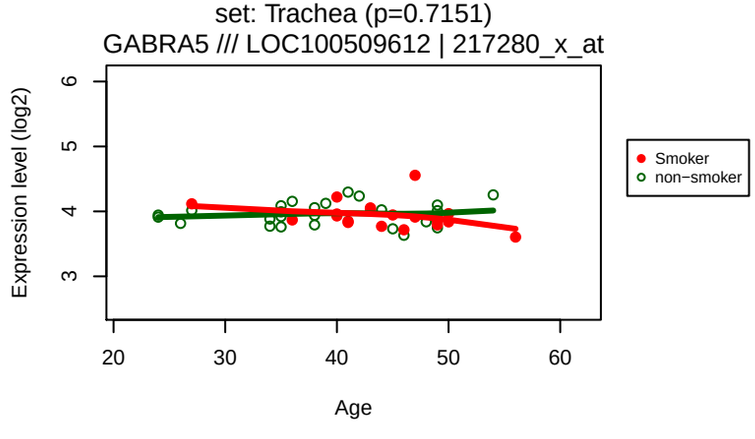

Gene expression ratio in Males
gene: GABRA5 /// LOC100509612 | 217280_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6845
p=0.5007
p=0.9303
p=0.2154
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GABRA5 /// LOC100509612 | 217280_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7595
p=0.6394
p=0.6141
p=0.9548
SmallAirway2
LargeAirway2
Trachea
Macrophage2

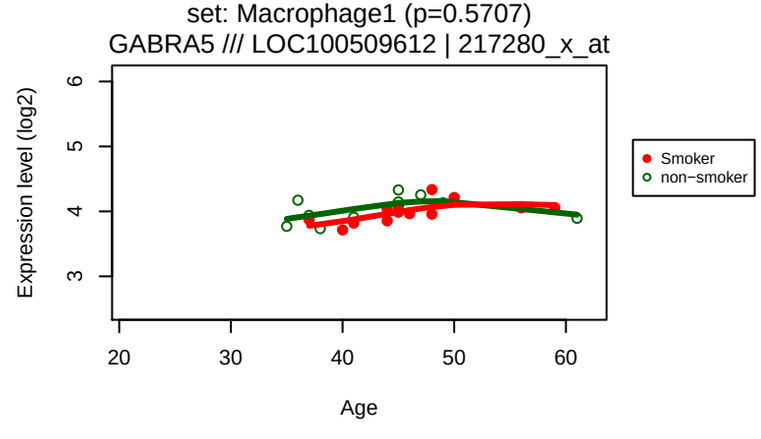

set: SmallAirway1 (p=0.7722)
GABRA5 /// LOC100509612 | 217280_x_at
Expression level (log2)
Age
Smoker
non−smoker

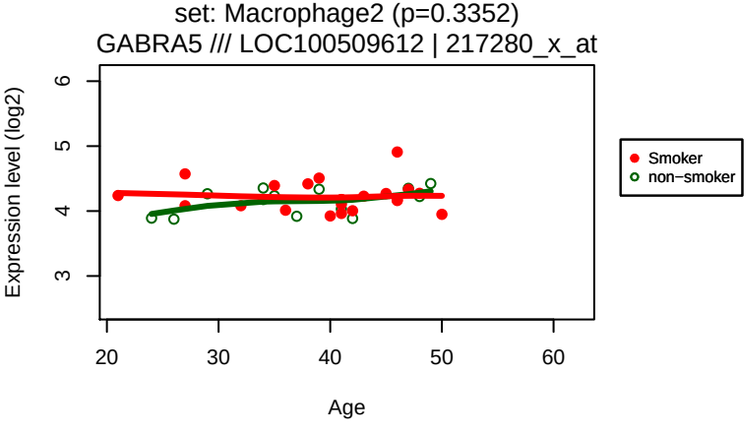

set: SmallAirway2 (p=0.7178)
GABRA5 /// LOC100509612 | 217280_x_at
Expression level (log2)
Age
Smoker
non−smoker

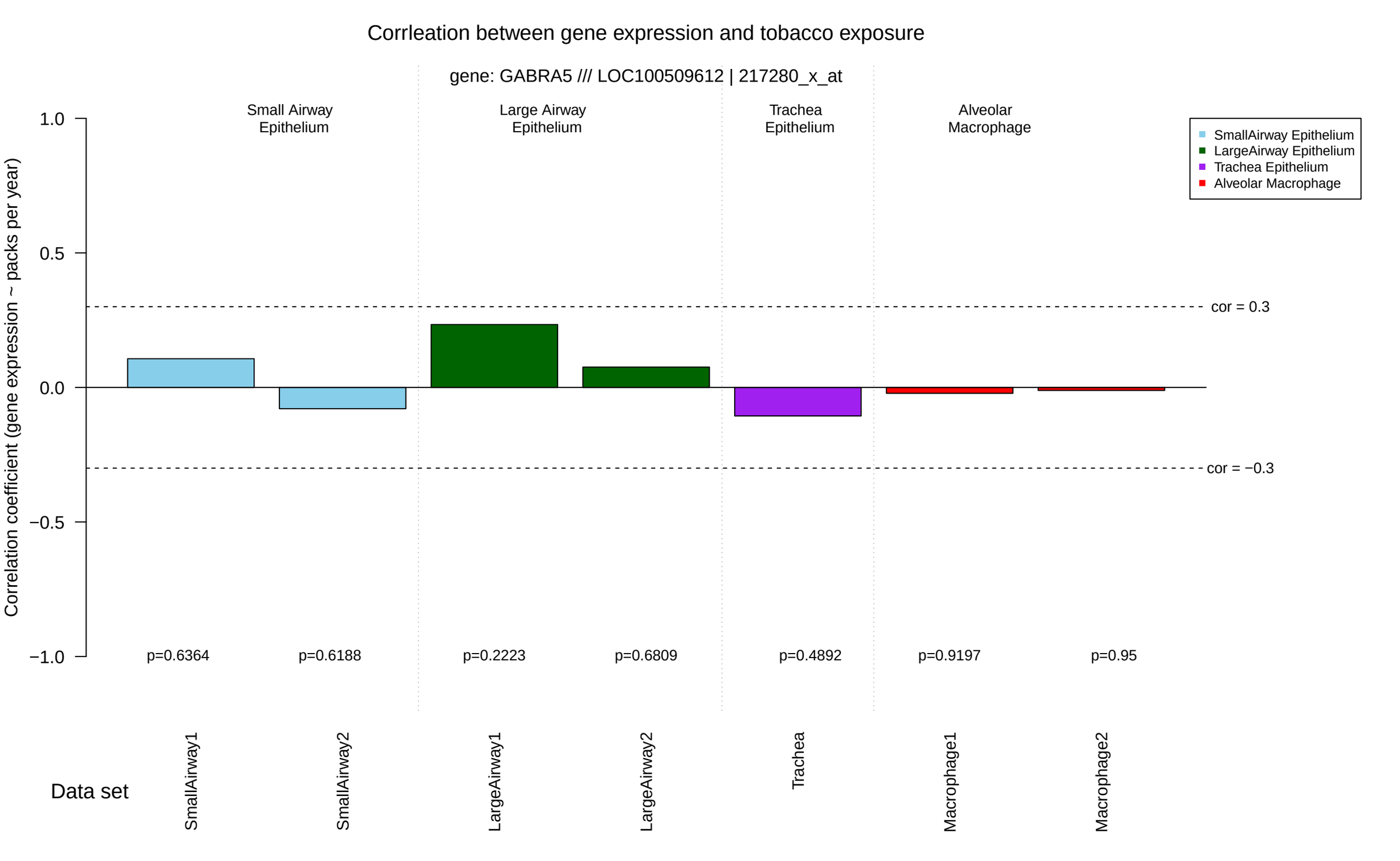

set: LargeAirway1 (p=0.0442)
GABRA5 /// LOC100509612 | 217280_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3829)
GABRA5 /// LOC100509612 | 217280_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7151)
GABRA5 /// LOC100509612 | 217280_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5707)
GABRA5 /// LOC100509612 | 217280_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3352)
GABRA5 /// LOC100509612 | 217280_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GABRA5 /// LOC100509612 | 217280_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6364
p=0.6188
p=0.2223
p=0.6809
p=0.4892
p=0.9197
p=0.95
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2